Elbert Chao

elbertchao.com | elbert.chao443@gmail.com | in/elbertc | github.com/elbertchao | 647-719-8849

Education

Western University — Sept. 2025 – Expected Jan 2028
Master of Engineering Science (MESc), Software Engineering — *London, ON*
GPA: 4.0/4.0
Thesis: Uncertainty Quantification for Deep Learning in Medical Imaging
Award: Med-i CREATE Competitive Scholarship ($10,000)
Coursework: Digital Image Processing, Dependable AI Systems, LLM/VLM, Data Analytics

Western University — Sept. 2021 – May 2025
Bachelor of Engineering Science (BESc), Software Engineering — *London, ON*
Honours: Dean's Honours List (4×) — **GPA:** 3.7/4.0
Coursework: Machine Learning, Artificial Intelligence I/II, Data Structures & Algorithms, Cloud Computing

Technical Skills

ML & AI Research: Medical Image Segmentation, CNNs, Transformers, Computer Vision
Frameworks & APIs: PyTorch, TensorFlow Lite, LangChain, OpenAI API, Streamlit
Backend & Data: FastAPI, Flask, MongoDB, REST APIs
Deployment & DevOps: GCP, Nginx, PM2
Languages: Python, TypeScript

Experience

Western University — Aug. 2025 – Present
Graduate Researcher (Python, PyTorch, CNNs, Transformers) — London, ON

- Improved kidney ablation segmentation by architecting **SR-MedT++**, a hybrid CNN–Transformer model with custom stratified resampling, achieving a **DSC/F1 score of 0.8165 $\pm$ 0.0224** and **IoU of 0.7082 $\pm$ 0.0257**.
- Produced pixel-level epistemic uncertainty maps using **Monte Carlo Dropout** and layer-wise heatmaps, enabling clinicians to identify low-confidence regions for targeted review before acting on model output.
- Built a scalable validation pipeline over **1,500 CT scans (125 patients)** to benchmark prediction variance and measure model reliability in the presence of real-world anomalies.

PM Accelerator — Jun. 2025 – Aug. 2025
AI Engineer Intern (Python, FastAPI, OpenAI API, MongoDB) — Remote (Boston, MA)

- Boosted resume-to-job alignment accuracy by **35%** by building a semantic matching system using **OpenAI API** embeddings and a **FastAPI** backend.
- Cut manual job data processing by **60%** by engineering a Python web scraping and text normalization pipeline to ingest and structure unstructured job listings at scale.
- Designed a scalable **RESTful API** to support resume parsing and context-aware interview question generation, reducing time-to-interview-prep for end users.

Free Appropriate Sustainability Technology – FAST — Nov. 2024 – Apr. 2025
Full-Stack Developer (Next, Python, Flask, Nginx, PM2) — London, ON

- Increased crop monitoring effectiveness by **90%** by building a full-stack ML platform — **Next.js (TypeScript)** frontend and **Flask (Python)** backend — that delivers real-time strawberry health insights to field users.
- Achieved classification speeds under **500ms** by systematically debugging edge cases and refining feature inputs through real-world model testing cycles.
- Sustained **99%+ uptime** over a 2-month live deployment with fewer than 3 total incidents by configuring **Nginx** for load balancing and **PM2** for process management and auto-recovery.
- Secured and configured a production environment for a **24/7** campus-deployed service, maintaining availability with zero configuration drift throughout the full academic term.

Projects

Warren Buffett Chatbot Agent — Jun. 2025 – Aug. 2025
AI Engineer (Python, OpenAI API, LangChain)

- Handled **100+** distinct financial query types with real-time retrieval by building an end-to-end conversational AI agent using **LangChain** and **OpenAI** models with multi-tool reasoning.
- Integrated external financial and search APIs to ground responses in live market data, enabling reliable, context-aware investment reasoning at query time.

FormFixer.AI — 4th Year Capstone Project — Sept. 2024 – Apr. 2025
Team Lead (TensorFlow Lite, Computer Vision)

- Reduced joint misclassification rates by **80%** during live video analysis by optimizing a **TensorFlow Lite** pose estimation model for real-time inference on-device.